



Course Code – MTH 302

Course Title – PROBABILITY AND STATISTICS

Zero Lecture



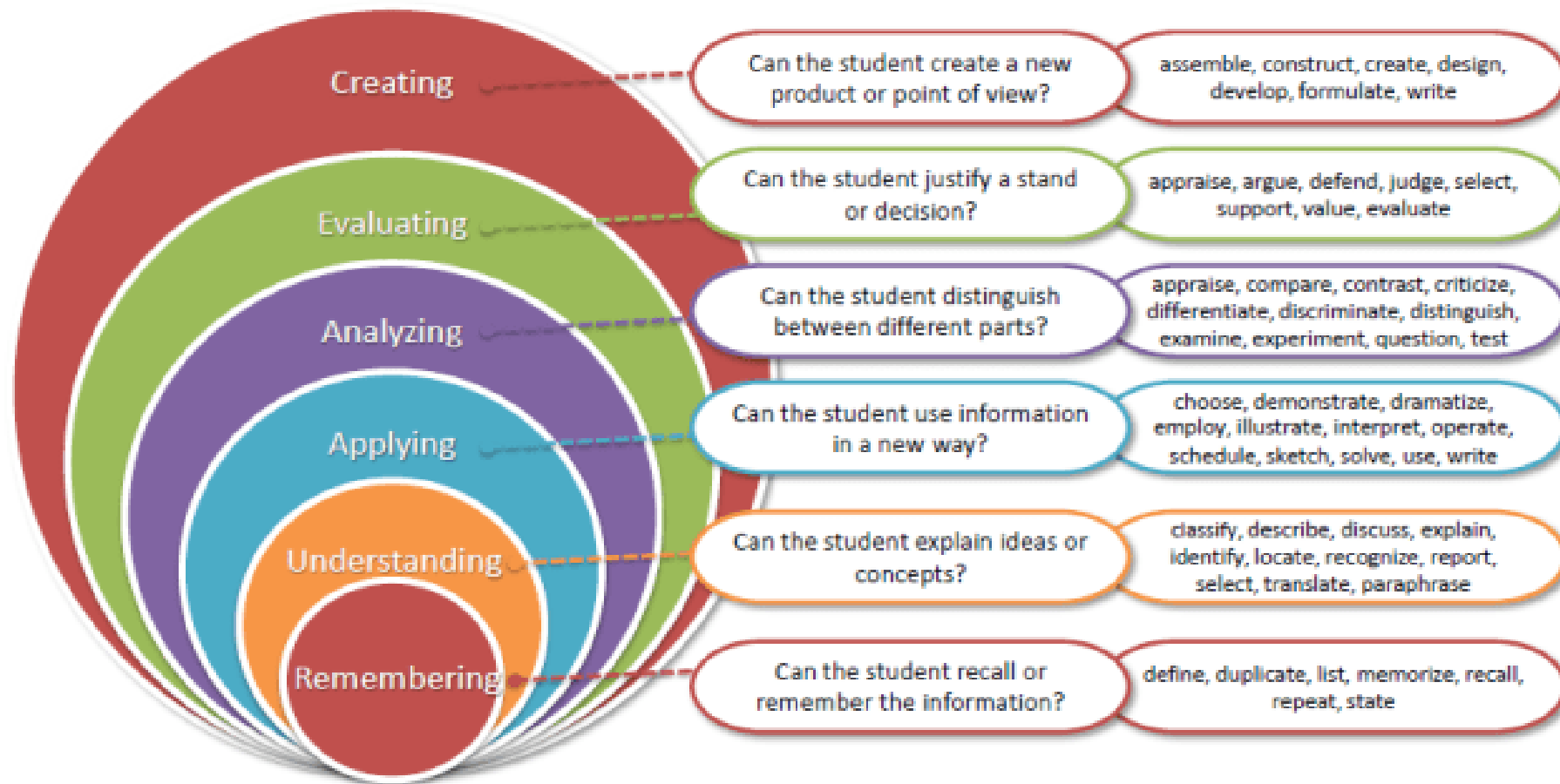
LTP and credit details

$L - T - P : 3 - 1 - 0$

(3 Lectures, 1 Tutorial, 0 Practical)

Credit:- 4

Bloom's Taxonomy



Course Outcomes

Through this course students should be able to

- recall the concept of random variables and its distribution functions.
- recognize relationships among the variables through correlation and regression.
- apply probability distributions to find the solution of different engineering problems.
- describe sample, population and statistical inference.
- understand hypothesis testing and its applications.

Programme Outcomes

- **PO 1:** Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization for the solution of complex engineering problems.
- **PO 2:** Problem analysis: Identify, formulate, research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- **PO 12:** Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.
- **PSO1::**Apply basic knowledge in the areas such as Software Engineering, Networking and Security, Database Management Systems, Intelligent Systems, Operating Systems, Programming Skills and System Architecture for building Software products along with the integration of futuristic technologies.

Course Focus Identification :

- **Employability**
- **Skill Development**
- **Data Visualization**

Unit Wise course contents:

Unit-1: Random Variables and Probability Distributions

- discrete and continuous random variables and their distribution functions
- Joint probability distributions
- mean, variance and covariance of random variables
- Chebyshev's theorem(without proof)

Unit Wise course contents:

Unit – 2 : Correlation and Linear regression

- scatter plots
- correlation coefficient and its properties
- Karl Pearson's correlation coefficient
- Spearman's rank correlation coefficient
- Linear regression and its properties

Unit Wise course contents:

Unit – 3 : Special Discrete Distributions

- Bernoulli process
- Binomial distribution
- Negative binomial distribution
- Geometric distribution
- Poisson distribution
- Moment generating function (without proof) of the above mentioned distributions.

Unit Wise course contents:

Unit – 4 : Special Continuous Distributions

- Normal distribution
- Gamma distribution
- Exponential distribution
- Moment generating function (without proof) of the above mentioned distributions.

Unit Wise course contents:

Unit – 5 : Point Estimation and the Central Limit Theorem

- the central limit theorem (without proof)
- unbiased estimator
- consistent estimator
- efficient and sufficient estimator
- maximum likelihood estimation

Unit Wise course contents:

Unit – 6 : Testing of Hypothesis

- types of error
- Z-test for single mean and difference of means
- Student's t-test for single mean and difference of means
- F-test
- Chi-square test for goodness of fit

Assessment and Evaluation Model

Bifurcation of Marks:

Attendance	5
CA (2 Best out of 3 subjective tests)	25
MTE (MCQ)	20
ETE (Mix Category)(Category:13)	50
Total	100



Academic Task details with Allotment/ Submission Week

Academic Task	Objective	Detail of Academic Task	Nature of Task	Mode of Academic Task	Max Marks	Allotment Week	Submission Week	Evaluation Parameters	Is Rubric applicable	Mapped Outcomes/ Add-Ones		
Test 1	To check the understanding of the students about random variables and its distribution functions	discrete and continuous random variables and their distribution functions, joint probability distributions, mean, variance and covariance of random variables, Chebyshev's theorem(without proof)	Individual	Offline	30	3	4	Applicable	No	View Outcomes View Add-Ones	Edit	Delete



Academic Task details with Allotment/ Submission Week

Academic Task	Objective	Detail of Academic Task	Nature of Task	Mode of Academic Task	Max Marks	Allotment Week	Submission Week	Evaluation Parameters	Is Rubric applicable	Mapped Outcomes/ Add-Ons		
Test 2	To test the understanding of students in correlation and linear regression and some discrete distributions.	scatter plots, correlation coefficient and its properties, Karl Pearson's correlation coefficient, Spearman's rank correlation coefficient, Linear regression and its properties, bernoulli process, binomial distribution, poisson distribution	Individual	Offline	30	5	6	Applicable	No	View Outcomes View Add-Ons	Edit	Delete



Academic Task details with Allotment/ Submission Week

Academic Task	Objective	Detail of Academic Task	Nature of Task	Mode of Academic Task	Max Marks	Allotment Week	Submission Week	Evaluation Parameters	Is Rubric applicable	Mapped Outcomes/ Add-Ones		
Test 3	To check the understanding and ability of the students to handle data and to apply suitable tool.	data will be provided to student and on the basis of the data student need to perform certain tasks to provide the answers to the questionnaire (It will be based on all the activities discussed till 11th week)	Individual	Offline	30	11	12	Applicable	No	View Outcomes View Add-Ones	Edit	Delete

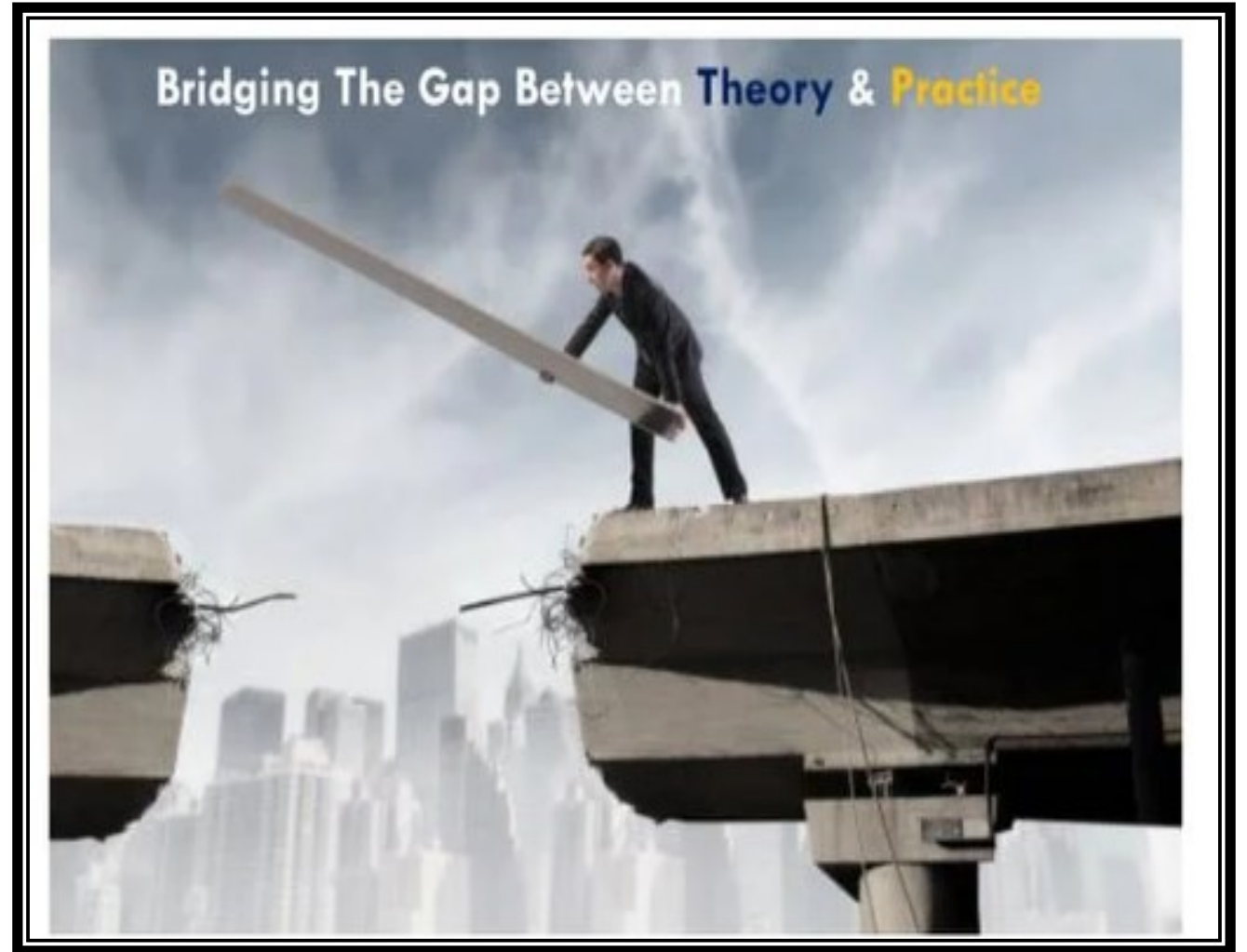


INNOVATIVE PEDAGOGIES (Add strategies)

INNOVATIVE IMPLEMENTATIONS IN PROBABILITY AND STATISTICS

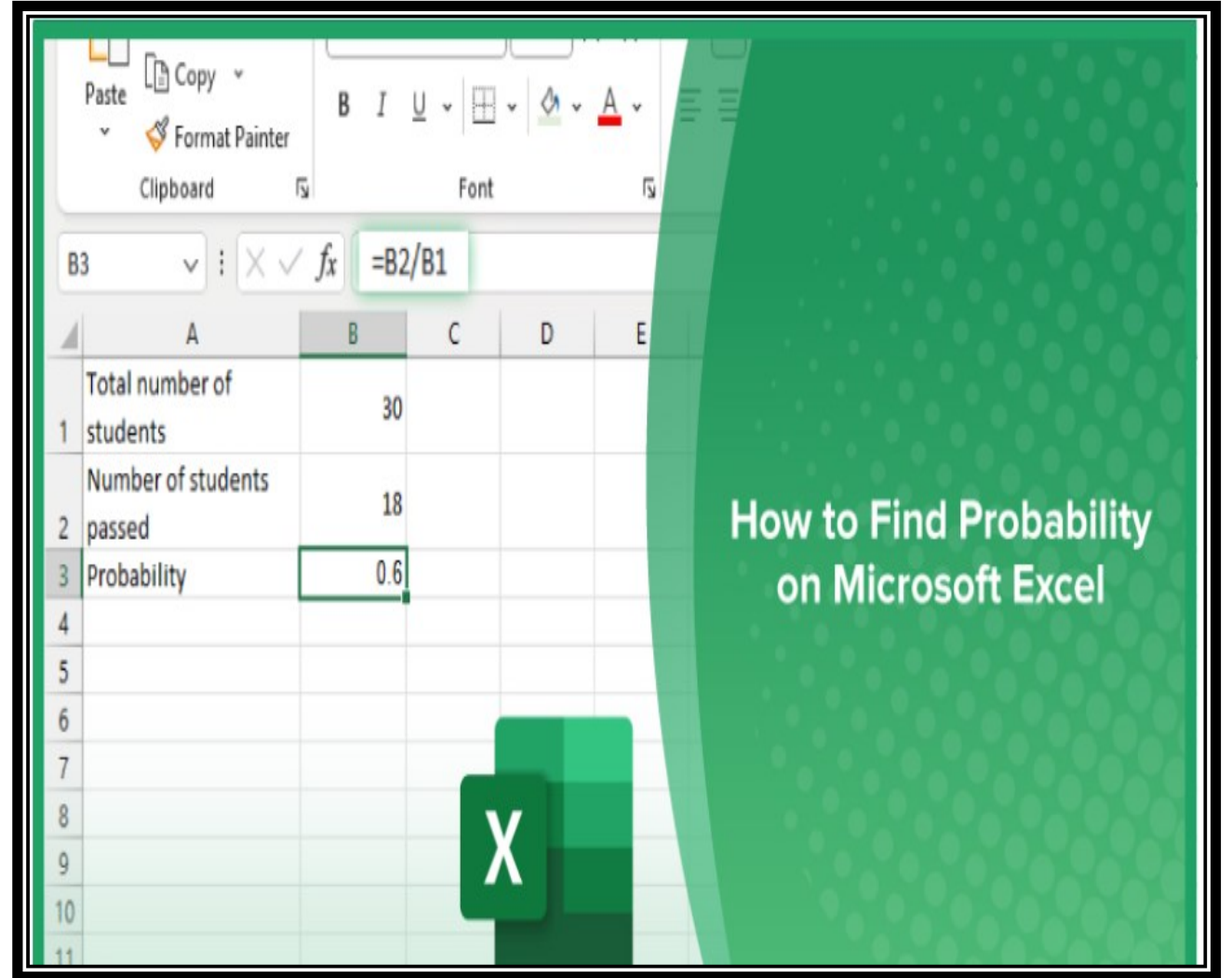
INNOVATIVE PEDAGOGIES (Bridging the Gap)

- **How to bridge the gap between Theoretical Probability and Statistics and its implementation in Excel?**



INNOVATIVE PEDAGOGIES (Probability and Microsoft Excel)

- **To explore Microsoft Excel from Probability Aspects!**



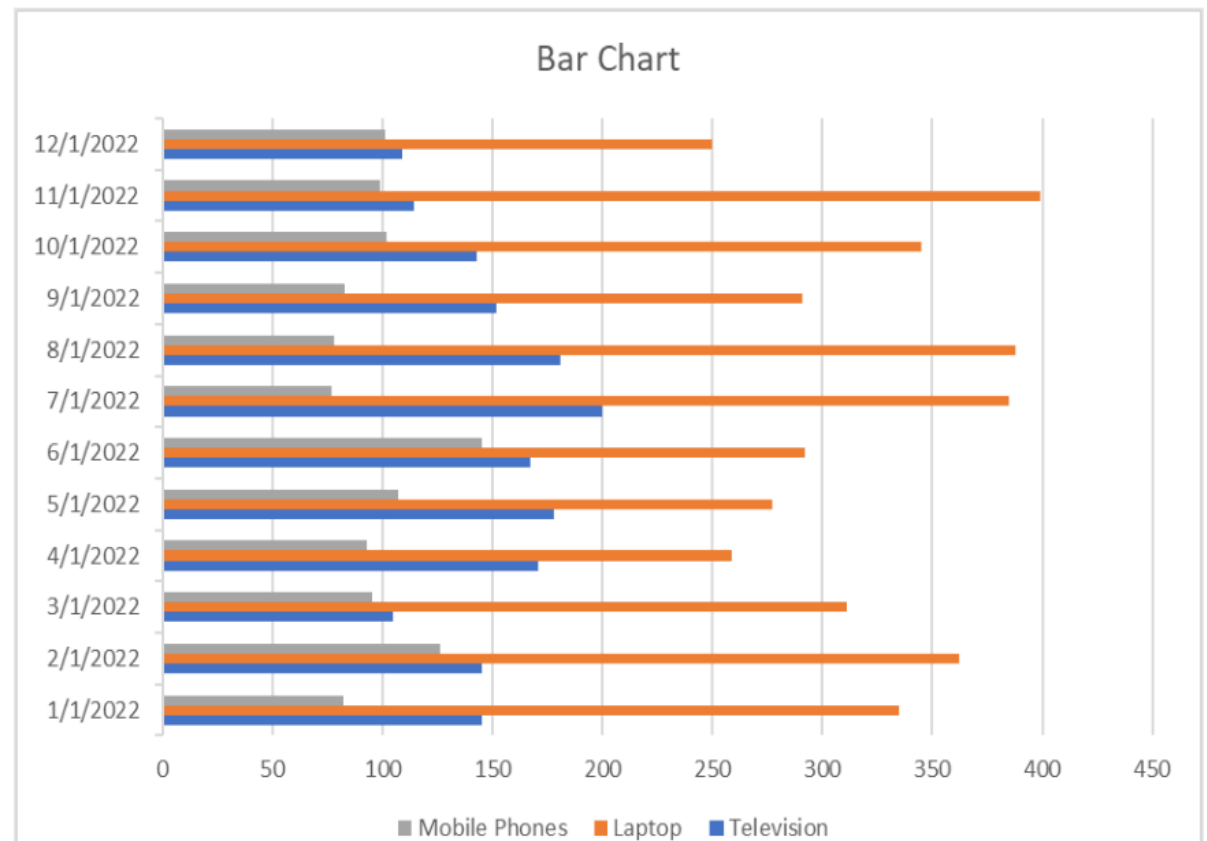
How to Find Probability on Microsoft Excel

	A	B	C	D	E
1	Total number of students	30			
2	Number of students passed	18			
3	Probability	0.6			
4					
5					
6					
7					
8					
9					
10					
11					

INNOVATIVE PEDAGOGIES (Data Visualization in Excel)

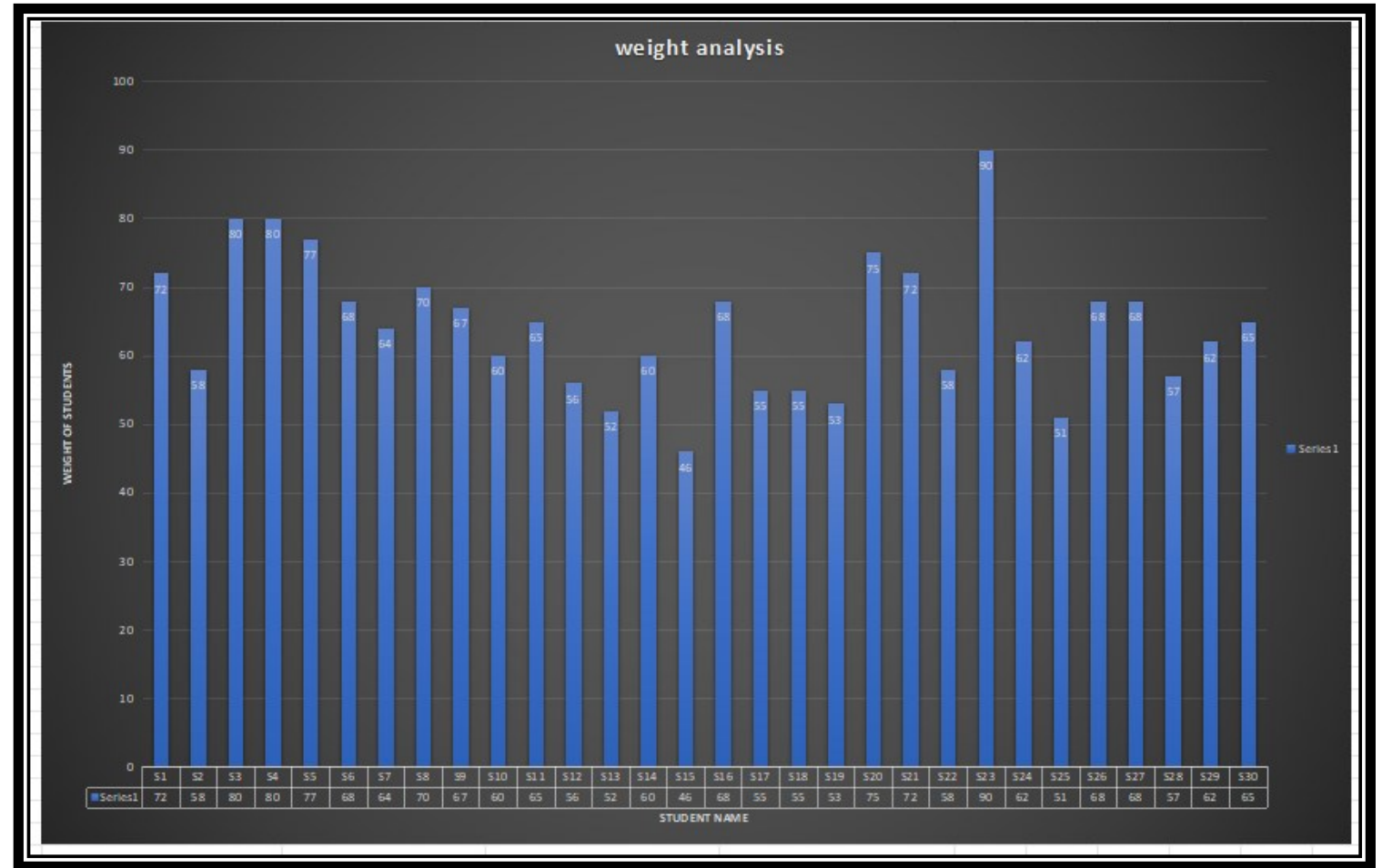
- Data Visualization in Excel
- Tabular Data in Charts? Yes it is possible!

	A	B	C	D
1	Month	Television	Laptop	Mobile Phones
2	1/1/2022	145	335	82
3	2/1/2022	145	362	126
4	3/1/2022	105	311	95
5	4/1/2022	171	259	93
6	5/1/2022	178	277	107
7	6/1/2022	167	292	145
8	7/1/2022	200	385	77
9	8/1/2022	181	388	78
10	9/1/2022	152	291	83
11	10/1/2022	143	345	102
12	11/1/2022	114	399	99
13	12/1/2022	109	250	101



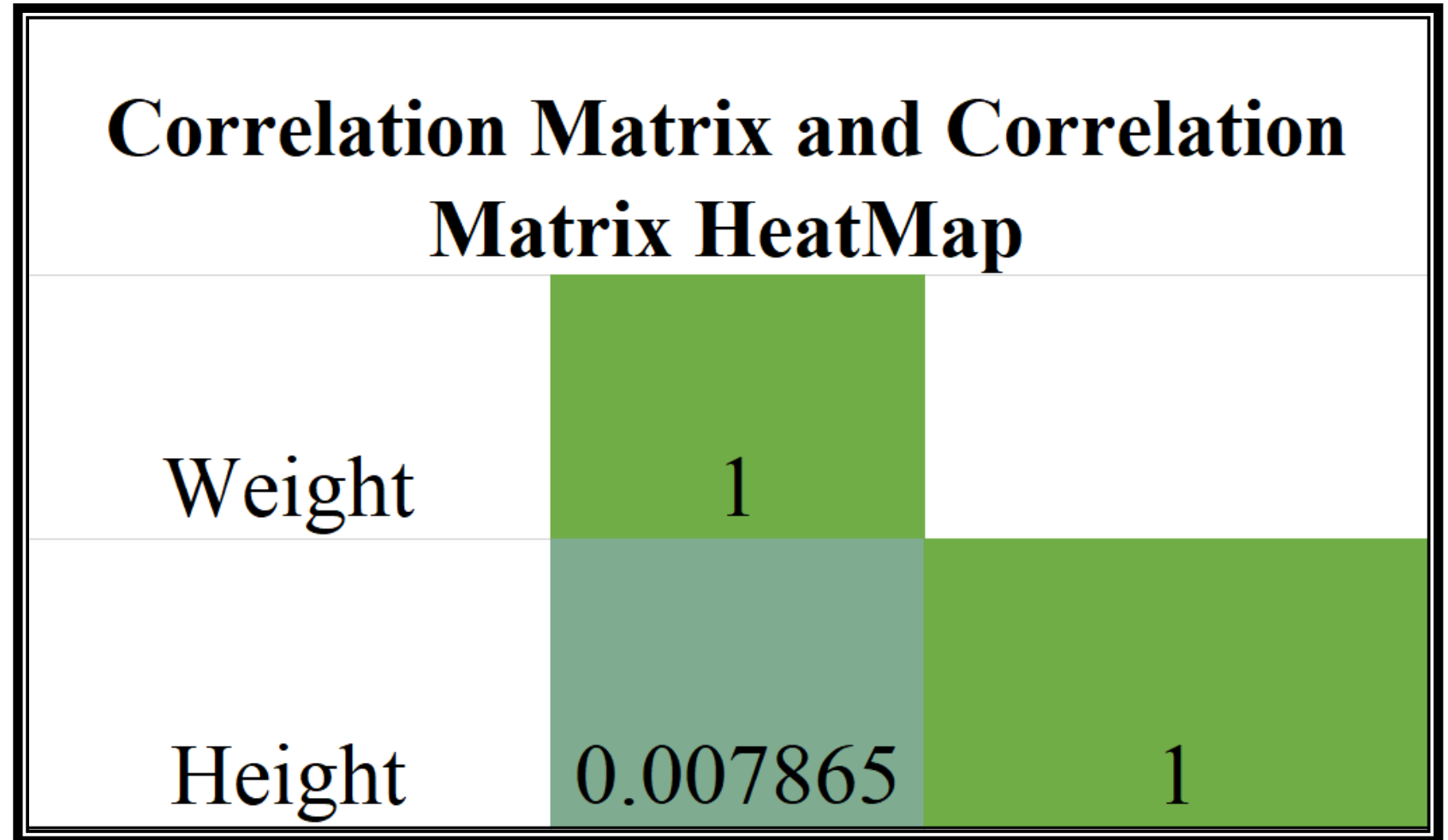
INNOVATIVE PEDAGOGIES (Raw data and its analysis)

- **Analysis of weight data collected in Excel.**



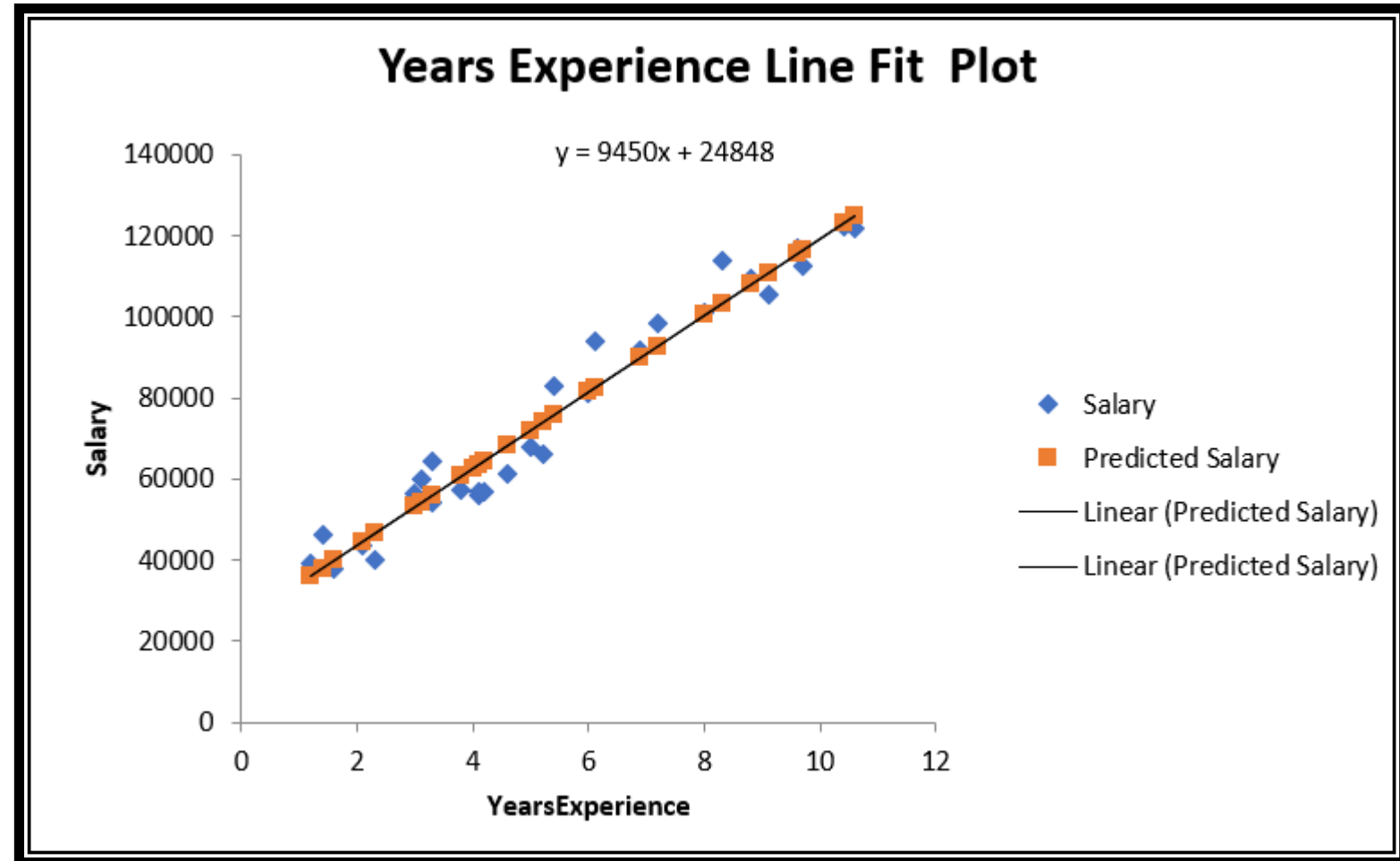
INNOVATIVE PEDAGOGIES (Correlation matrix heatmap)

- **Correlation Matrix and Correlation Matrix Heat Map**



INNOVATIVE PEDAGOGIES (Regression analysis)

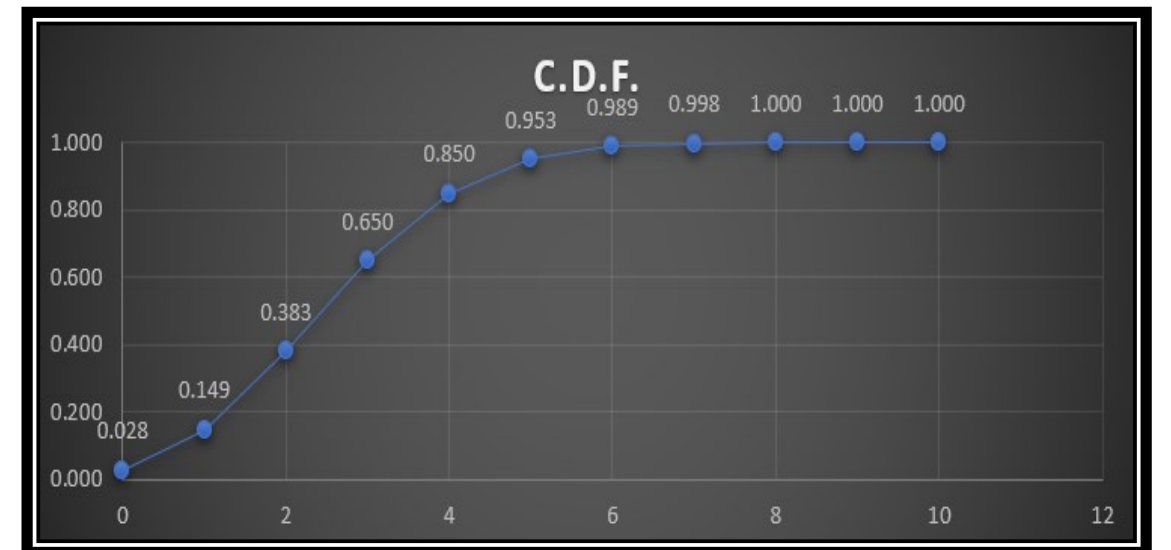
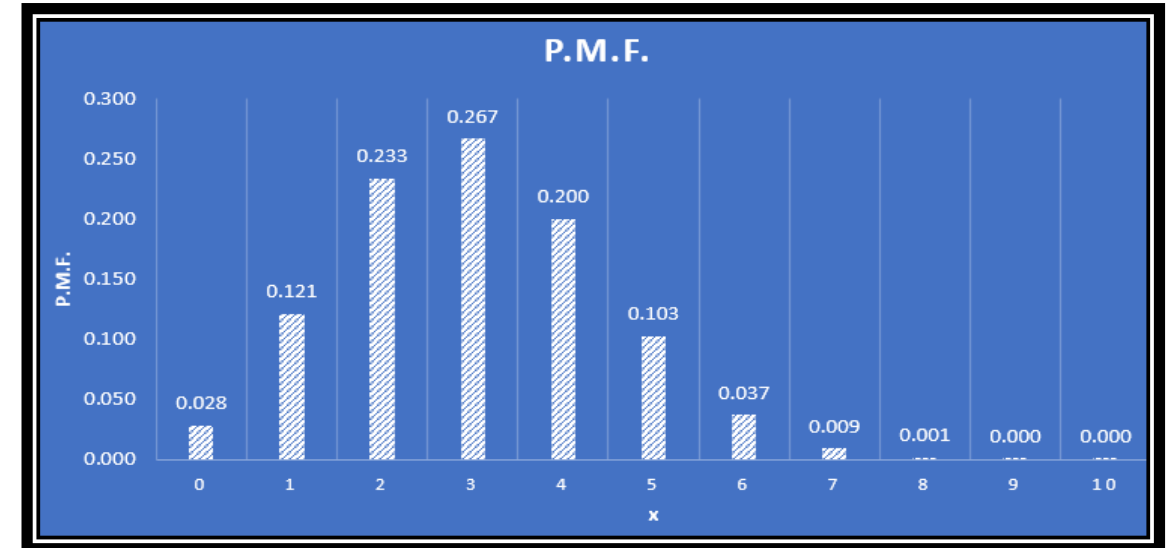
- **Regression Analysis Practical Implementation**



INNOVATIVE PEDAGOGIES (Binomial distribution with proper analysis)

• Binomial Distribution – A Novel Analysis Aspect

$X = x$	$P[X = x]$ [P.M.F.]	$P[X \leq x]$ [Cumulative Distribution Function]
0	0.028	0.028
1	0.121	0.149
2	0.233	0.383
3	0.267	0.650
4	0.200	0.917
5	0.103	0.989
6	0.037	0.998
7	0.009	0.999
8	0.001	1.000
9	0.000	1.000
10	0.000	1.000



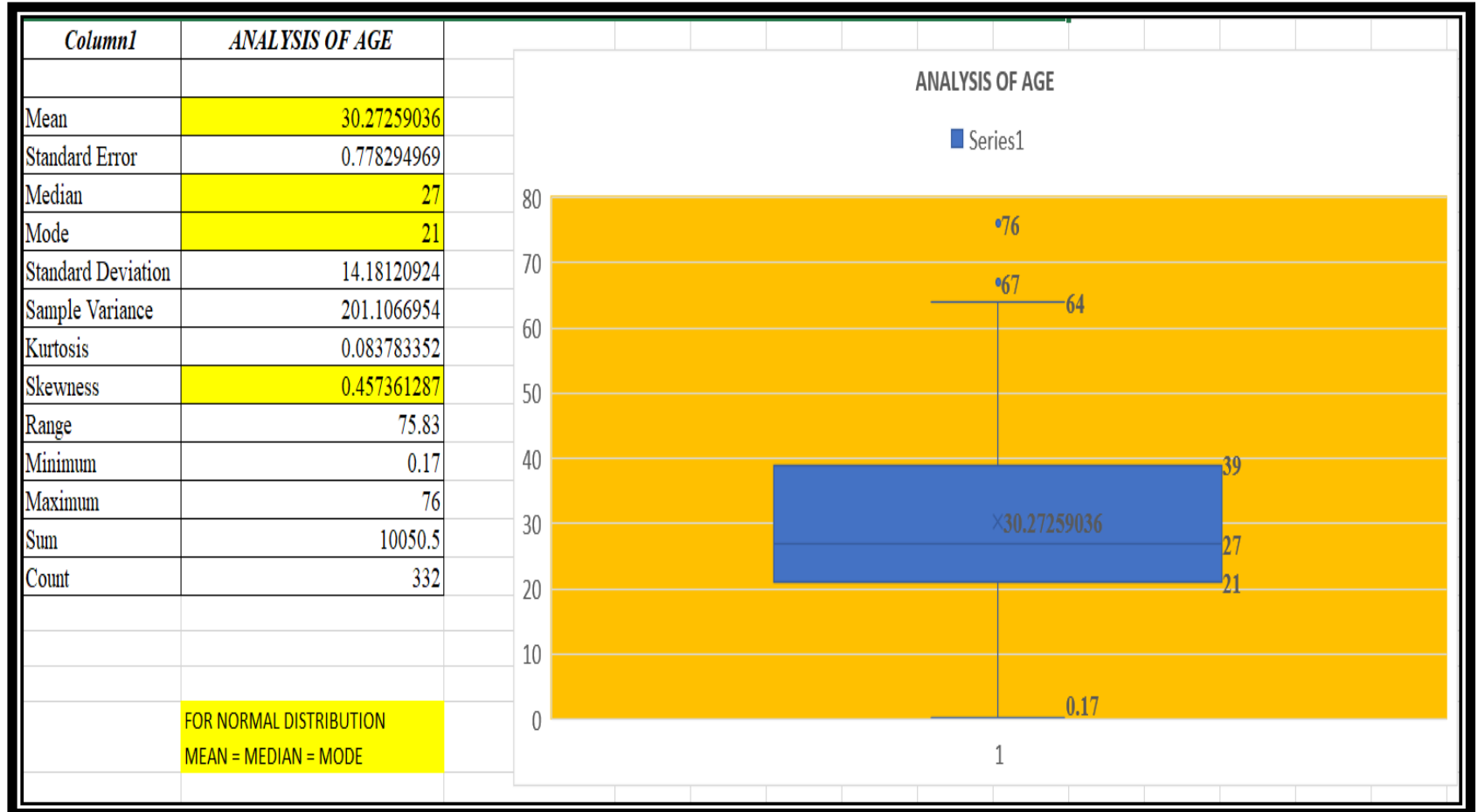
INNOVATIVE PEDAGOGIES (Paired t test via Excel)

- Analysis and Validity of analysis in Medical Field via Statistical Test
- Paired t test

Paired t test					
BEFORE INSULIN DATA OF SUGAR LEVEL	AFTER INSULIN DATA OF SUGAR LEVEL	Null Hypothesis : There is no effect of giving insulin to the patients Mean 1 = Mean 2	Alternate Hypothesis : There is an effect of Insulin on the blood sugar levels Mean 1 > Mean 2 [Right tailed]	level of significance = 0.05	degrees of freedom = 10 - 1 = 9
310	208				
315	215	t-Test: Paired Two Sample for Means			
320	218				
325	225		<i>Variable 1</i>	<i>Variable 2</i>	
330	230	Mean	331.1	212	
335	235	Variance	178.7666667	1245.333333	
340	240	Observations	10	10	
345	143	Pearson Correlation	-0.137996804		
350	250	Hypothesized Mean Difference	0		
341	156	df	9		
		t Stat	9.553032019		
		P(T<=t) one-tail	2.61431E-06		
		t Critical one-tail	1.833112933		
		P(T<=t) two-tail	5.22862E-06		
		t Critical two-tail	2.262157163		

INNOVATIVE PEDAGOGIES (Identification of Normal distribution)

- Is the data Normally Distributed or Not?



INNOVATIVE PEDAGOGIES (Analysis of Chi square test)

Chi- square test

- **How to check significance of difference between Observed frequencies and expected frequencies?**

The screenshot shows an Excel spreadsheet with the following data:

Observed Frequencies					
	Type of defect				
Shift	A	B	C	D	Row Total
1	15	21	45	13	
2	26	31	34	5	
3	33	17	49	20	
Co	=SUM(B4:B6)				

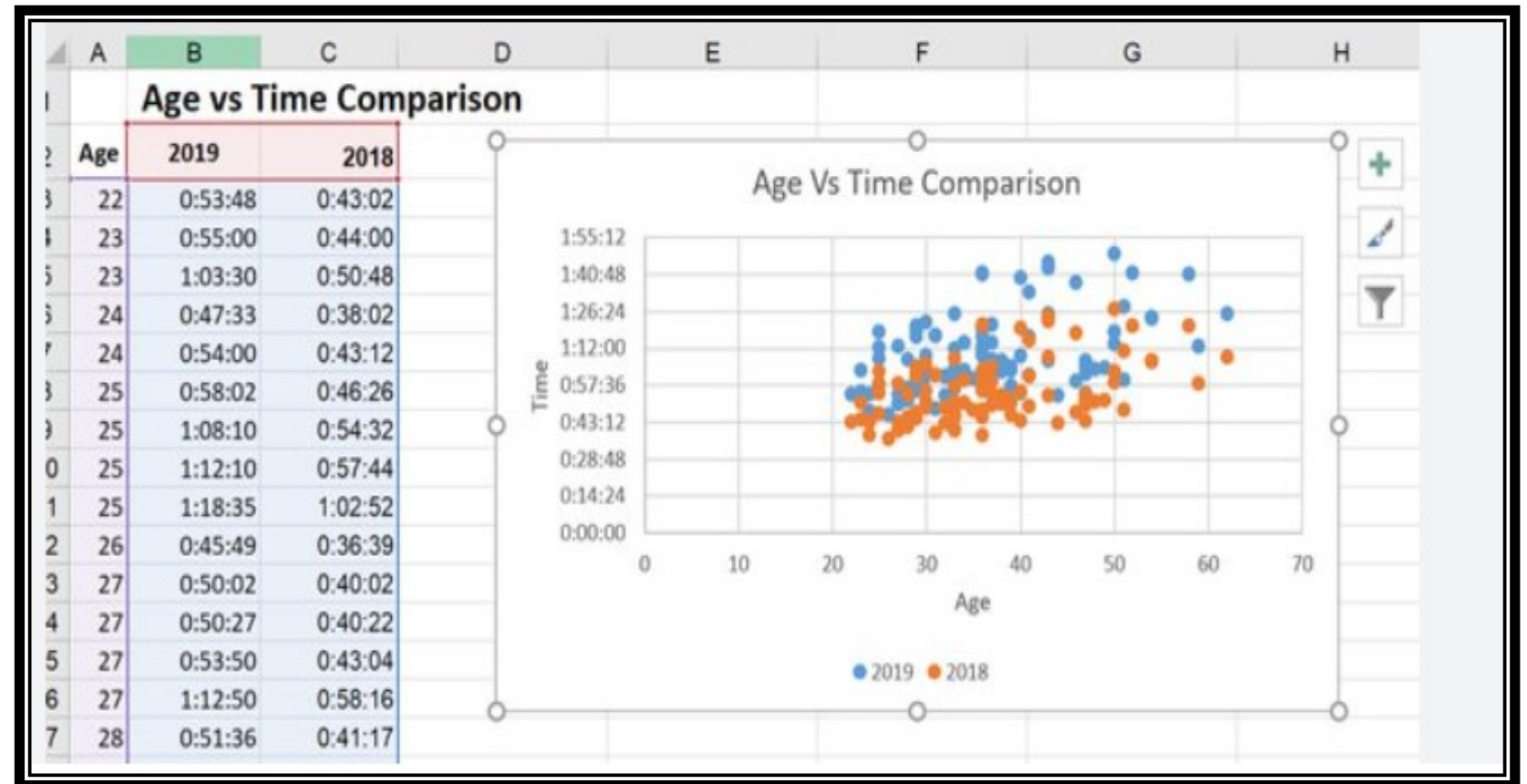
Expected Frequencies				
	Type of defect			
Shift	A	B	C	D
1				
2				
3				

The formula bar shows the formula `=SUM(B4:B6)` entered in cell B7.

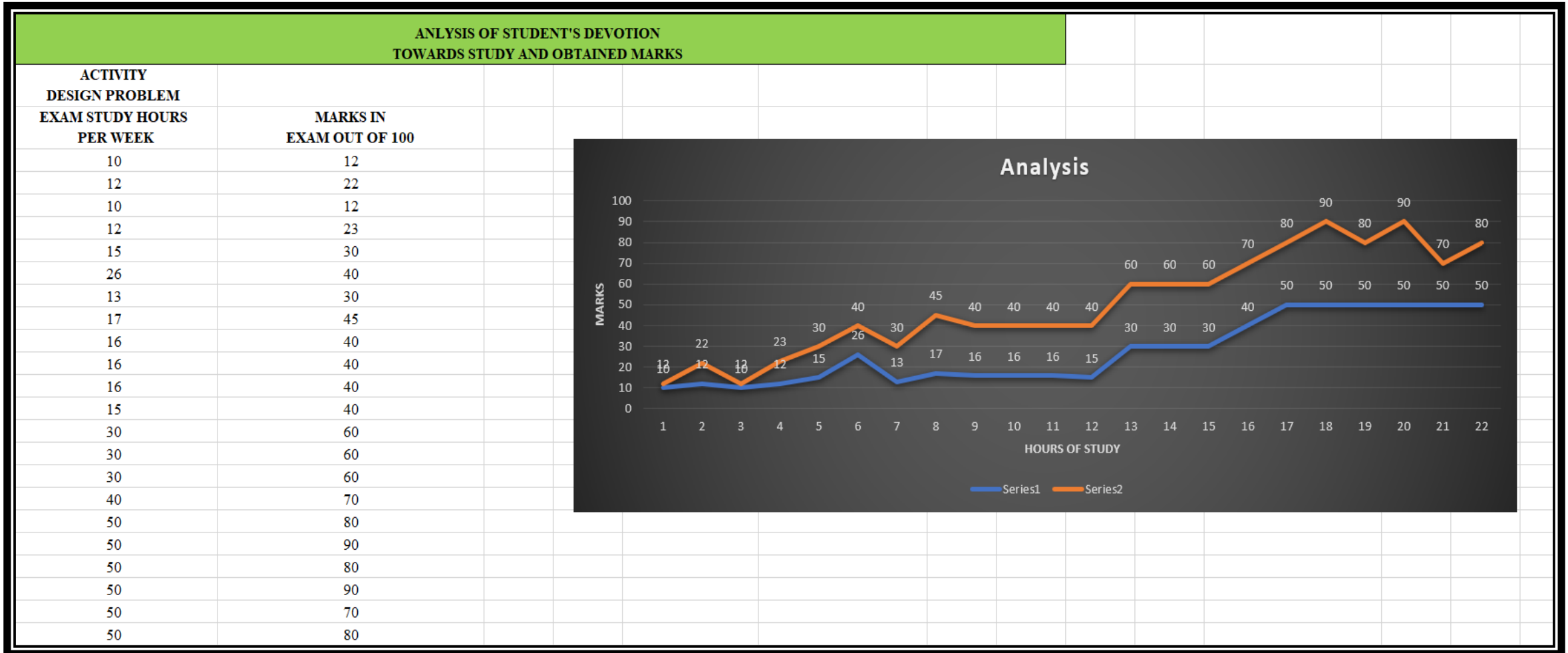
INNOVATIVE PEDAGOGIES (Scatter plots in Excel)

Scatter Plots

- Can you identify type of Correlation from the provided image?

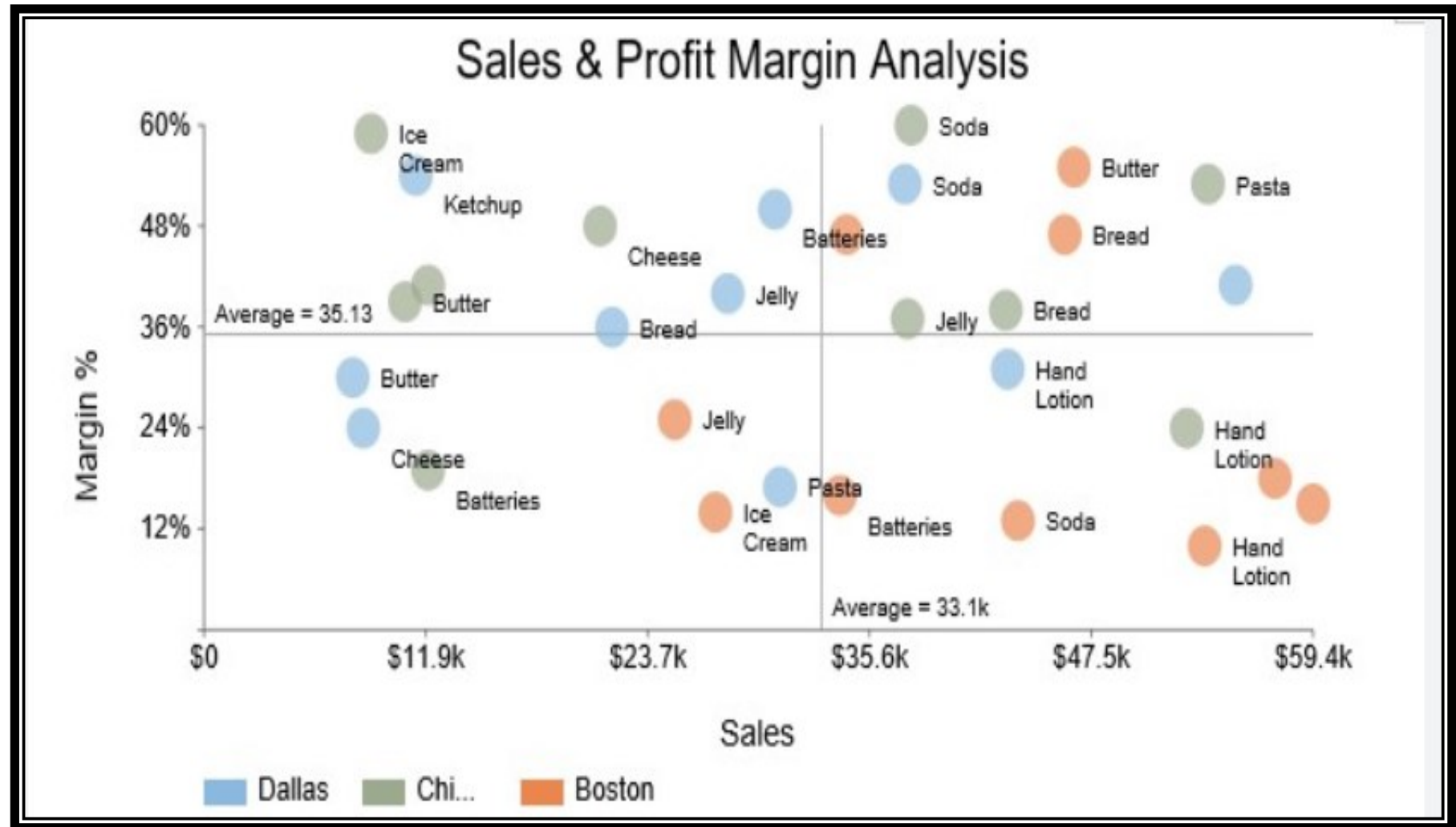


INNOVATIVE PEDAGOGIES (Analysis of different parameters)



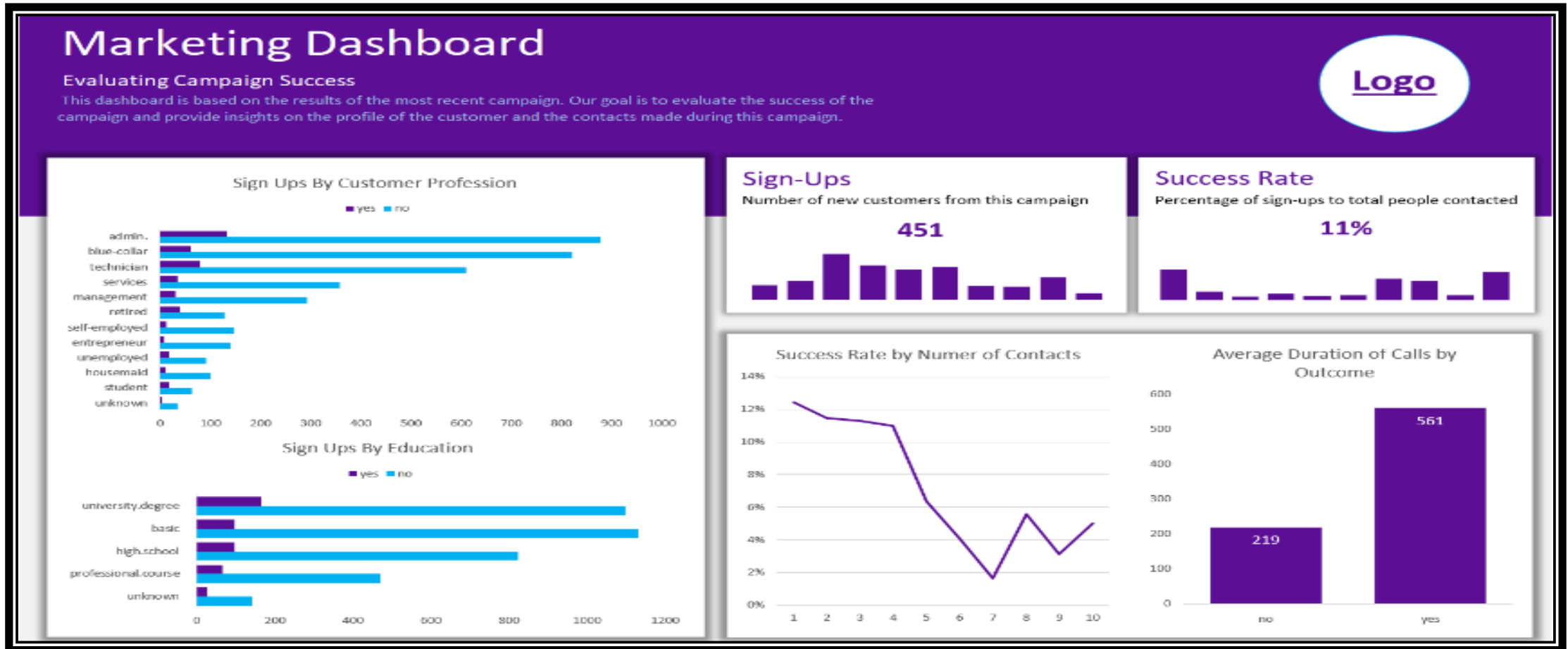
INNOVATIVE PEDAGOGIES (Analysis of different parameters)

- Sales and Profit Margin Analysis via Scatter Plot?**



INNOVATIVE PEDAGOGIES (Marketing Dashboard)

Dashboards in Excel? Possible or Not? Yes, it is possible.



INNOVATIVE PEDAGOGIES (Sales Dashboard)



Books Required

Text Book:

- “PROBABILITY AND STATISTICS FOR ENGINEERS AND SCIENTISTS” by RONALD E. WALPOLE, RAYMOND H. MYERS, SHARON L. MYERS, AND KEYING YE, PEARSON

References Books:

- “PROBABILITY AND STATISTICS FOR ENGINEERS” by Dr. J. Ravichandran [WILEY]

Link - <https://www.wileyindia.com/probability-and-statistics-for-engineers-as-per-aicte.html>

- “FUNDAMENTALS OF MATHEMATICAL STATISTICS” by S.C.GUPTA AND V.K.KAPOOR, SULTAN CHAND & SONS (P) LTD.

MOOC or Industry certification associated with the course

Probability Theory for Data Science

By Prof. Ishapathik Das | IIT Tirupati

Link of MOOC is:

https://onlinecourses.nptel.ac.in/noc24_ma64/preview

Summary

Course Status :	Ongoing
Course Type :	Core
Duration :	12 weeks
Category :	◦ Mathematics
Credit Points :	3
Level :	Undergraduate
Start Date :	22 Jul 2024
End Date :	11 Oct 2024
Enrollment Ends :	05 Aug 2024
Exam Registration Ends :	16 Aug 2024
Exam Date :	27 Oct 2024 IST

Note: This exam date is subjected to change based on seat availability. You can check final exam date on your hall ticket.



Unit mapped	Broad topic	Sub Topic	Source Type	Source Title	*%age mapping (approx)	Source URL
Unit 1	Random Variables and Probability Distributions	discrete and continuous random variables and their distribution functions, joint probability distributions, mean, variance and covariance of random variables, Chebyshev's theorem(without proof)	NPTEL Reading Material	Random variables and its distributions	70%(approx)	https://archive.nptel.ac.in/courses/111/104/111104032/
Unit 2	Correlation and Linear regression	scatter plots, correlation coefficient and its properties, Karl Pearson's correlation coefficient, Spearman's rank correlation coefficient, Linear regression and its properties	Reading Material	Lecture Notes	70%(approx)	https://ebooks.inflibnet.ac.in/mgmt15/chapter/ccorrelation-karl-pearsons-coefficient-of-correlation-spearman-rank-correlation-ar/
Unit 3	Special Discrete Distributions	bernoulli process, binomial distribution, negative binomial distribution, geometric distribution, poisson distribution, moment generating function (without proof) of the above mentioned distributions.	Reading Material	Special distributions	40%(approx.)	https://www.probabilitycourse.com/chapter3/3_1_5_special_discrete_distr.php
Unit 4	Special Continuous Distributions	normal distribution, gamma distribution, exponential distribution, moment generating function (without proof) of the above mentioned distributions.	NPTEL Course	Probability Theory for Data Science	80% (approx)	https://onlinecourses.nptel.ac.in/noc24_ma64/preview
Unit 5	Point Estimation and the Central Limit Theorem	the central limit theorem(without proof), unbiased estimator, consistent estimator, efficient and sufficient estimator, maximum likelihood estimation	Reading Material	Introduction to Estimation	80%(approx.)	Unit-5.pdf (egyankosh.ac.in)
Unit 6	Testing of Hypothesis	types of error, Z-test for single mean and difference of means, Student's t- test for single mean and difference of means, F- test, Chi-square test for goodness of fit	NPTEL Video	Introduction to Statistical Hypothesis Testing	60%(approx.)	https://archive.nptel.ac.in/courses/103/106/103106120/



INNOVATIVE PEDAGOGIES

PROBABILITY AND STATISTICS CONCEPTS WITH IMPLEMENTATIONS AND APPLICATIONS

INNOVATIVE PEDAGOGIES (Random Numbers)

Generating Random Numbers

- **RAND()**: Generates a random number between 0 and 1.
- **RANDBETWEEN(bottom, top)**: Generates a random integer between the specified bottom and top values.

INNOVATIVE PEDAGOGIES (Descriptive Statistics)

Descriptive Statistics

- **AVERAGE(range)**: Calculates the mean of a set of numbers.
- **MEDIAN(range)**: Finds the median value.
- **MODE.SNGL(range)**: Determines the mode.
- **STDEV.P(range)**: Computes the standard deviation for a population.
- **STDEV.S(range)**: Computes the standard deviation for a sample.
- **VAR.P(range)**: Calculates the variance for a population.
- **VAR.S(range)**: Calculates the variance for a sample.

INNOVATIVE PEDAGOGIES (Probability Distributions)

Probability Distributions

- **NORM.DIST(x, mean, standard_dev, cumulative)**: Returns the normal distribution.
- **NORM.INV(probability, mean, standard_dev)**: Returns the inverse of the normal cumulative distribution.
- **BINOM.DIST(number_s, trials, probability_s, cumulative)**: Returns the binomial distribution.
- **POISSON.DIST(x, mean, cumulative)**: Returns the Poisson distribution.
- **EXPON.DIST(x, lambda, cumulative)**: Returns the exponential distribution.

INNOVATIVE PEDAGOGIES (Hypothesis Testing)

Hypothesis Testing

- **T.TEST(array1, array2, tails, type)**: Performs a T-test to determine if two samples are likely to have come from the same two underlying populations that have the same mean.
- **Z.TEST(array, x, sigma)**: Returns the one-tailed P-value of a Z-test.
- **CHISQ.TEST(actual_range, expected_range)**: Returns the test for independence.

INNOVATIVE PEDAGOGIES (Correlation and Regression)

Correlation and Regression

- **CORREL(array1, array2)**: Calculates the correlation coefficient between two arrays.
- **SLOPE(known_y's, known_x's)**: Returns the slope of the linear regression line.
- **INTERCEPT(known_y's, known_x's)**: Returns the intercept of the linear regression line.
- **LINEST(known_y's, [known_x's], [const], [stats])**: Returns the statistics for a line by using the least squares method.

INNOVATIVE PEDAGOGIES

Data Analysis Toolpak (Add-In)

- **Random Number Generation:** Generates random numbers from different distributions.
- **Descriptive Statistics:** Provides a summary of descriptive statistics for a dataset.
- **Correlation:** Computes the correlation matrix for a range of data.
- **Regression:** Performs linear regression analysis.

INNOVATIVE PEDAGOGIES (Applications related to Unit 1 content)

Unit I: Random Variables and Probability Distributions Applications

- 1.Finance:** Modeling stock prices using discrete and continuous random variables.
- 2.Insurance:** Calculating premiums based on risk assessments using joint probability distributions.
- 3.Quality Control:** Using distribution functions to monitor and control manufacturing processes.

INNOVATIVE PEDAGOGIES (Applications related to Unit 2)

Unit II: Correlation and Linear Regression Applications

- 1. Market Analysis:** Analyzing the relationship between sales and marketing spend using correlation and regression.
- 2. Health Studies:** Investigating the correlation between lifestyle factors and health outcomes.
- 3. Real Estate:** Predicting property prices based on various features like location, size, and amenities.

INNOVATIVE PEDAGOGIES (Application related to Unit 3 content)

Unit III: Special Discrete Distributions Applications

- 1.Customer Service:** Modeling the number of customer service calls using a Poisson distribution.
- 2.Quality Assurance:** Applying binomial and negative binomial distributions in defect testing.
- 3.Queueing Theory:** Using geometric distribution to model waiting times in queues.

INNOVATIVE PEDAGOGIES (Applications related to Unit 4 content)

Unit IV: Special Continuous Distributions Applications

- 1. Reliability Engineering:** Using exponential distribution to model time-to-failure of components.
- 2. Environmental Science:** Modeling rainfall patterns using gamma distribution.
- 3. Finance:** Risk assessment using normal distribution for returns on investment.

INNOVATIVE PEDAGOGIES (Applications related to Unit 5 content)

Unit V: Point Estimation and the Central Limit

Theorem

Applications

- 1.Survey Analysis:** Estimating population parameters from survey samples.
- 2.Market Research:** Using point estimation to analyze market trends.
- 3.Quality Control:** Applying the central limit theorem to improve sampling techniques in manufacturing.

INNOVATIVE PEDAGOGIES (Applications related to Unit 6 content)

Unit VI: Testing of Hypothesis Applications

1.Clinical Trials: Testing the effectiveness of new drugs using hypothesis testing.

2.A/B Testing: Comparing different versions of a product to determine which performs better.

3.Educational Research: Evaluating the impact of new teaching methods using hypothesis tests.



INNOVATIVE PEDAGOGIES

LETS TALK ABOUT SOME
POSSIBLE PROJECT WORKS

INNOVATIVE PEDAGOGIES (Project related to Unit 1)

Unit I: Random Variables and Probability Distributions

Risk Assessment in Insurance: Create a model to assess risk and calculate premiums for insurance policies.

INNOVATIVE PEDAGOGIES (Project related to Unit 2)

Unit II: Correlation and Linear Regression

Health Outcome Study: Study the correlation between different lifestyle choices (e.g., diet, exercise) and health outcomes.

INNOVATIVE PEDAGOGIES (Project related to Unit 3)

Unit III: Special Discrete Distributions

Defect Testing in Manufacturing: Use binomial and negative binomial distributions to model defect rates in a production line.

INNOVATIVE PEDAGOGIES (Project related to Unit 4)

Unit IV: Special Continuous Distributions

Risk Assessment in Investment: Develop a model to assess investment risks using normal distribution.

INNOVATIVE PEDAGOGIES (Project related to Unit 5)

Unit V: Point Estimation and the Central Limit Theorem

Market Trend Analysis: Analyze market research data to estimate trends and make predictions.

INNOVATIVE PEDAGOGIES (Project related to Unit 6)

Unit VI: Testing of Hypothesis

Teaching Method Evaluation: Collect and analyze data to evaluate the impact of different teaching methods on student performance.



Cohorts: Data Analyst, Business Analyst, Statistician

Skill Attainment

- **Skill Development**
- **Employability based on Statistical analysis**
- **Data Visualization Expert**

Library e-resources

Library e-resources :(One path is given)

Reading Material on the course:

Path:

UMS Main Page



LIBRARY SEARCH



USEFUL LINKS



NATIONAL DIGITAL LIBRARY: ENTER YOUR TOPIC

